Math 248 Midterm

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Problem 1. Let $n \in \mathbb{N}$ and I be a non-empty open interval in $\mathbb{R}$. Let $A : I \to M(n, \mathbb{C})$ and $B : M(n, \mathbb{C})$ be smooth paths.

1. Prove that $[A(t)]' = [A'(t)]^T$.

Proof. To show the above result, we need to show that

$$(A^T)'_{ij}(t) = (A')^T_{ij}(t) \ \ \forall 1 \leq i, j \leq n.$$

Since $(A^T)_{ij} = A_{ji}$, we obtain that

$$\begin{aligned}
(A^T)'_{ij} &= \lim_{\Delta t \to 0} \frac{(A^T)_{ij}(t + \Delta t) - (A^T)_{ij}}{\Delta t} \\
&= \lim_{\Delta t \to 0} \frac{A_{ji}(t + \Delta t) - A_{ji}(t)}{\Delta t} \\
&= A'_{ji}(t) \\
&= (A_{ij})^T(t)
\end{aligned}$$

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2. Prove that $[A(t)B(t)]' = A'(t)B(t) + A(t)B'(t)$.

Proof. We have that

$$\begin{aligned}
[A(t)B(t)]' &= \lim_{\Delta t \to 0} \frac{A(t + \Delta t)B(t + \Delta t) - A(t)B(t)}{\Delta t} \\
&= \lim_{\Delta t \to 0} \Big[B(t + \Delta t) \frac{A(t + \Delta t) - A(t)}{\Delta t} + A(t) \frac{B(t + \Delta t) - B(t)}{\Delta t} \Big] \\
&= B(t) \lim_{\Delta t \to 0} \frac{A(t + \Delta t) - A(t)}{\Delta t} + A(t) \lim_{\Delta t \to 0} \frac{B(t + \Delta t) - B(t)}{\Delta t} \\
&= B(t)A'(t) + A(t)B'(t).
\end{aligned}$$

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3. Prove that if, in addition, $A(t)$ is invertible for all t, then

$$[A^{-1}(t)]' = -A^{-1}(t)A'(t)A^{-1}(t).$$

Proof. Note that since $A(t)$ is invertible for all t, it follows that $AA^{-1} = I$. Using part (2), we have $[A(t)A^{-1}(t)]' = 0$. Differentiating, we see that

$$\begin{aligned}
A'(t)A^{-1}(t) + A(t)[A^{-1}(t)]' = 0 &\implies A(t)[A^{-1}(t)]' = -A'(t)A^{-1}(t) \\
&\implies (A^{-1})'(t) = A^{-1}(t)(-A'(t))A^{-1}(t).
\end{aligned}$$

That is,

$$\begin{aligned}(A^{-1})'(t) &= A^{-1}(t)(-A'(t))A^{-1}(t)\\ &= -A^{-1}(t)A'(t)A^{-1}(t).\end{aligned}$$

Hence, we conclude that

$$(A^{-1})'(t) = -A^{-1}(t)A'(t)A^{-1}(t).$$

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Problem 2. 1. Explain why $E = [3, 5)$ is a closed and bounded subset of the metric space $X = (1, 5)$.

2. Explain why $E = [3, 5)$ is not a compact subset of the metric space $X = (1, 5)$.

3. Explain why the following argument is faulty. "$O(n)$ is compact because it is a closed subset of $\mathrm{GL}(n, \mathbb{R})$ and is bounded."

Solution. 1. E is closed since $[3, 5) = [3, 6] \cap (1, 5)$ where $[3, 6]$ is closed and bounded set in $\mathbb{R}$.

2. $E = [3, 5)$ is not compact in X because E is NOT closed in $\mathbb{R}$ (that is, $5 \notin E$). Hence, E cannot be compact in $X = (1, 5)$ by fact 1.

3. This argument is faulty because in general if $X = \mathbb{R}^k$, then we cannot guarantee that a set being closed and bounded in X will be compact in X. In particular, $O(n)$ being closed and bounded in $\mathrm{GL}(n, \mathbb{R})$ does not imply that it is a compact set in $\mathrm{GL}(n, \mathbb{R})$ since $\mathrm{GL}(n, \mathbb{R}) \neq \mathbb{R}^k$. Furthermore, $\mathrm{GL}(n, \mathbb{R})$ is not even a compoact set in $\mathbb{R}^{n^2}$ as we have proven in class and so the fact

"closed subsets of compact sets are compact"

cannot be used to make such a statement.

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Problem 3. Let V be an $n-$dimensional vector space. Suppose $T : V \to V$ is a linear transformation. As we know, once we equip V with a basis, we can represent T by a matrix. Let $\mathcal{B}$ and $\mathcal{C}$ be two bases for V. Denote the matrix of T relative to the basis $\mathcal{B}$ by $A_{\mathcal{B}}$ and the matrix of T relative to the basis $\mathcal{C}$ by $A_{\mathcal{C}}$. Thus, for every $\mathbf{x} \in V$, we have

$$\begin{aligned}&[T(\mathbf{x})]_{\mathcal{B}} A_{\mathcal{B}}[\mathbf{x}]_{\mathcal{B}},\\ &[T(\mathbf{x})]_{\mathcal{C}} = A_{\mathcal{C}}[\mathbf{x}]_{\mathcal{C}}.\end{aligned}$$

Prove that

$$\mathcal{A}_{\mathcal{C}} = P_{\mathcal{C}\leftarrow\mathcal{B}}\mathcal{A}_{\mathcal{B}}\Big(P_{\mathcal{C}\leftarrow\mathcal{B}}\Big)^{-1}.$$

Proof. Let $[\mathbf{x}]_{\mathcal{C}} \in \mathbb{R}^n$ be arbitrary. Then we have

$$\begin{aligned}P_{\mathcal{C}\leftarrow\mathcal{B}}A_{\mathcal{B}}\Big((P_{\mathcal{C}\leftarrow\mathcal{B}})^{-1}[\mathbf{x}]_{\mathcal{C}}\Big) &= P_{\mathcal{C}\leftarrow\mathcal{B}}\mathcal{A}_{\mathcal{B}}\Big(P_{\mathcal{B}\leftarrow\mathcal{C}}[\mathbf{x}]_{\mathcal{C}}\Big)\\ &= P_{\mathcal{C}\leftarrow\mathcal{B}}(\mathcal{A}_{\mathcal{B}}[\mathbf{x}]_{\mathcal{B}})\\ &= P_{\mathcal{C}\leftarrow\mathcal{B}}[T(\mathbf{x})]_{\mathcal{B}}\\ &= [T(\mathbf{x})]_{\mathcal{C}}\\ &= \mathcal{A}_{\mathcal{C}}[\mathbf{x}]_{\mathcal{C}}.\end{aligned}$$

Hence, we have

$$P_{\mathcal{C}\leftarrow\mathcal{B}}\mathcal{A}_{\mathcal{B}}\Big((P_{\mathcal{C}\leftarrow\mathcal{B}})^{-1}[\mathbf{x}]_{\mathcal{C}}\Big) = \mathcal{A}_{\mathcal{C}}[\mathbf{x}]_{\mathcal{C}}.$$

Multiplying $[\mathbf{x}]_{\mathcal{C}}^{-1}$ on both sides, we obtain that

$$\mathcal{A}_{\mathcal{C}} = P_{\mathcal{C}\leftarrow\mathcal{B}}\mathcal{A}_{\mathcal{B}}(P_{\mathcal{C}\leftarrow\mathcal{B}})^{-1}.$$

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Problem 4. Let $N = \begin{pmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}$

1. Prove that N is nilpotent.

Proof. We have

$$\begin{aligned} N^2 &= \begin{pmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}^2 \\ &= \begin{pmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ &= O. \end{aligned}$$

Thus, N is nilpotent with $k = 2$ such that $N^2 = O$. ■

2. Compute $\exp(N)$.

Solution. Note that for all $k \geq 2$, $A^k = O$. Using the definition of the matrix exponential, we obtain

$$\begin{aligned} \exp(N) = \sum_{m=0}^{\infty} \frac{N^m}{m!} &= \frac{N^0}{0!} + \frac{N^1}{1!} + \frac{N^2}{2!} + 0 + 0 + \cdots \\ &= I + N \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 0 & -1 & -1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{pmatrix}. \end{aligned}$$

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Problem 5. Let $\theta \in \mathbb{R}$. Find the exponential of the matrix $X = \begin{pmatrix} 0 & -\theta & 0 \\ \theta & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$.

Solution. Let $\lambda_1 = i\theta$, $\lambda_2 = -i\theta$ and $\lambda_3 = 0$. Observe that

$$x\begin{pmatrix}1\\i\\0\end{pmatrix} = \begin{pmatrix}0 & -\theta & 0\\ \theta & 0 & 0\\ 0 & 0 & 0\end{pmatrix}\begin{pmatrix}1\\i\\0\end{pmatrix} = \begin{pmatrix}-i\theta\\ \theta\\ 0\end{pmatrix} = i\theta\begin{pmatrix}1\\i\\0\end{pmatrix}$$
$$x\begin{pmatrix}i\\1\\0\end{pmatrix} = \begin{pmatrix}0 & -\theta & 0\\ \theta & 0 & 0\\ 0 & 0 & 0\end{pmatrix}\begin{pmatrix}i\\1\\0\end{pmatrix} = -i\theta\begin{pmatrix}-\theta\\ i\theta\\ 0\end{pmatrix} = -i\theta\begin{pmatrix}i\\1\\0\end{pmatrix}$$
$$x\begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{pmatrix}0 & -\theta & 0\\ \theta & 0 & 0\\ 0 & 0 & 0\end{pmatrix}\begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix} = 0\begin{pmatrix}0\\0\\1\end{pmatrix}.$$

Then we see that $P = \begin{pmatrix}1 & i & 0\\ i & 1 & 0\\ 0 & 0 & 1\end{pmatrix}$ and $P^{-1} = \begin{pmatrix}1/2 & i/2 & 0\\ -i/2 & 1/2 & 0\\ 0 & 0 & 1\end{pmatrix}$. Furthermore, we obtain

$$\begin{aligned}
P^{-1}DP &= \begin{pmatrix}1 & i & 0\\ i & 1 & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}-i\theta & 0 & 0\\ 0 & i\theta & 0\\ 0 & 0 & 0\end{pmatrix}\begin{pmatrix}1/2 & i/2 & 0\\ -i/2 & 1/2 & 0\\ 0 & 0 & 1\end{pmatrix}\\
&= \begin{pmatrix}-i\theta & -\theta & 0\\ \theta & i\theta & 0\\ 0 & 0 & 0\end{pmatrix}\begin{pmatrix}1/2 & -i/2 & 0\\ -i/2 & 1/2 & 0\\ 0 & 0 & 1\end{pmatrix}\\
&= \begin{pmatrix}0 & -\theta & 0\\ \theta & 0 & 0\\ 0 & 0 & 0\end{pmatrix}\\
&= X.
\end{aligned}$$

Hence, we have

$$X = P^{-1}DP \text{ where } D = \begin{pmatrix}-i\theta & 0 & 0\\ 0 & i\theta & 0\\ 0 & 0 & 0\end{pmatrix}.$$

Since $e^X = Pe^DP^{-1}$, we obtain that

$$\begin{aligned}
e^X &= Pe^DP^{-1}\\
&= \begin{pmatrix}i & 1 & 0\\ 1 & i & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}e^{-i\theta} & 0 & 0\\ 0 & e^{i\theta} & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}1/2 & -i/2 & 0\\ -i/2 & 1/2 & 0\\ 0 & 0 & 1\end{pmatrix}\\
&= \begin{pmatrix}ie^{-i\theta} & e^{i\theta} & 0\\ e^{-i\theta} & ie^{i\theta} & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}1/2 & -i/2 & 0\\ -i/2 & 1/2 & 0\\ 0 & 0 & 1\end{pmatrix}\\
&= \begin{pmatrix}\frac{e^{-i\theta}-e^{i\theta}}{2} & \frac{-ie^{-i\theta}+ie^{i\theta}}{2} & 0\\ \frac{ie^{-i\theta}-ie^{i\theta}}{2} & \frac{e^{-i\theta}+e^{i\theta}}{2} & 0\\ 0 & 0 & 1\end{pmatrix}\\
&= \begin{pmatrix}\frac{e^{-i\theta}-e^{i\theta}}{2} & \frac{-(ie^{i\theta}-e^{-i\theta})}{2i} & 0\\ \frac{e^{i\theta}-e^{-i\theta}}{2i} & \frac{e^{-i\theta}+e^{i\theta}}{2} & 0\\ 0 & 0 & 1\end{pmatrix}\\
&= \begin{pmatrix}\cos\theta & -\sin\theta & 0\\ \sin\theta & \cos\theta & 0\\ 0 & 0 & 1\end{pmatrix}.
\end{aligned}$$

Hence, we have that

$$e^X = Pe^D P^{-1} = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

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Problem 6. Let $A \in O(n)$ and $B \in M(n, \mathbb{R})$. Prove that $\exp(ABA^T) = A\exp(B)A^T$.

Proof. One can prove via induction that $\exp(ABA^T) = \sum_{m=0}^{\infty} \frac{(ABA^T)^m}{m!}$. Using this fact, we obtain

$$\begin{aligned} \exp(ABA^T) = \sum_{m=0}^{\infty} \frac{(ABA^T)^m}{m!} &= \sum_{m=0}^{\infty} \frac{AB^m A^T}{m!} \\ &= \sum_{m=0}^{\infty} A\frac{B^m}{m!}A^T \\ &= A\sum_{m=0}^{\infty} \frac{B^m}{m!}A^T \\ &= A\exp(B)A^T. \end{aligned}$$

Now, we conclude that

$$\exp(ABA^T) = A\exp(B)A^T.$$

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Problem 7. Let G be any of the matrix Lie groups that we have introduced as embedded submanifolds of $\mathrm{GL}(n, \mathbb{F})$. Prove that $T_I G$ is a linear subspace of $\mathbb{R}^k$ where $k = n^2$ if $\mathbb{F} = \mathbb{R}$ and $k = 2n^2$ if $\mathbb{F} = \mathbb{C}$.

Proof. It suffices to show that $T_I G$ is closed under addition and scalar multiplication.

(i) $T_I G$ is closed under addition.

Let $X, Y \in T_I G$. Then there exists smooth paths $f_1(t), f_2(t) \in G$ such that $f_1(0) = I$, $f_1'(0) = X$, $f_2(0) = I$, and $f_2'(0) = X$. Define $f_3(t) = f_1(t)f_2(t)$. Then notice that

$$f_3'(t) = f_1'(t)f_2(t) + f_2'(t)f_1(t)$$

and so

$$\begin{aligned} f_3'(0) &= f_1'(0)f_2(0) + f_2'(0)f_1(0) \\ &= XI + YI \\ &= X + Y. \end{aligned}$$

Thus, $f_3'(0) = X + Y$ and clearly $f_3(0) = I$ since

$$f_3(0) = f_1(0)f_2(0) = II = I.$$

Therefore, we have $X + Y \in T_I G$.

(ii) $T_I G$ is closed under scalar multiplication.

Let $\alpha \in \mathbb{F}$ and let $X \in T_I G$. Then there exists a smooth path $f(t)$ in G such that $f(0) = I$ and $f'(0) = X$. Define $g(t) = f(\alpha t)$. Using the chain rule, we get

$$g'(t) = f'(\alpha t) \cdot \alpha \Longrightarrow g'(0) = \alpha \cdot f'(0) = \alpha X$$

and $g(0) = f(\alpha \cdot 0) = f(0) = I$. Hence, we see that $\alpha X \in T_I G$.

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Problem 8. Let $\mathbb{F}$ denote either $\mathbb{R}$ or $\mathbb{C}$. Prove that the *commutator bracket* $[A, B] = AB - BA$ defines a Lie Bracket n the vector space $M(n, \mathbb{F})$. (Thus, $M(n, \mathbb{F})$ equipped with this Lie Bracket is a Lie algebra, often denoted by $\mathfrak{gl}(n, \mathbb{F})$).

Proof. (i) Let $\alpha, \lambda \in \mathbb{F}$ and let $A, B, C \in M(n, \mathbb{F})$. Then

$$\begin{aligned}
[\alpha A + \lambda B, C] &= (\alpha A + \lambda B)C - C(\alpha A + \lambda B) \\
&= \alpha(AC) + \lambda(BC) - C(\alpha A) - C(\lambda B) \\
&= \alpha(AC) + \lambda(BC) - \alpha(CA) - \lambda(CB) \\
&= \alpha[AC - CA] + \lambda[BC - CB] \\
&= \alpha[A, C] + \lambda[B, C]
\end{aligned}$$

Thus, we obtain

$$[\alpha A + \lambda B, C] = \alpha[A, C] + \lambda[B, C].$$

A similar argument shows linearity in the second component.

(ii) Observe that

$$[A, B] = AB - BA = -(BA - AB) = -[B, A].$$

(iii) Let $X, Y, Z \in M(n, \mathbb{F})$. Note that

$$\begin{aligned}
[X, [Y, Z]] &= X[Y, Z] - [Y, Z]X \\
&= X(YZ - ZY) - (YZ - ZY)X \\
&= X(YZ) - X(ZY) - (YZ)X + (ZY)X
\end{aligned}$$

Then we have

$$\begin{aligned}
[Y, [Z, X]] &= Y(ZX) - Y(XZ) - (ZX)Y + (XZ)Y \\
[Z, [X, Y]] &= Z(XY) - Z(YX) - (XY)Z + (YX)Z.
\end{aligned}$$

Then we have

$$\begin{aligned}
&[X, [Y, Z]] + [Z, [X, Y]] + [Y, [Z, X]] \\
&= X(YZ) - (XY)Z + (YX)Z - Y(XZ) + (XZ)Y - X(ZY) \\
&+ Y(ZX) - (YZ)X + (ZY)X - Z(YX) + (ZX)Y - (ZX)Y \\
&= 0.
\end{aligned}$$

Hence, $[\cdot, \cdot]$ satisfies the Jacobi Identity. Thus, we conclude that $[A, B] = AB - BA$ defines a Lie Bracket on $M(n, \mathbb{F})$ where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

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Problem 9. Let V be a finite-dimensional real or complex vector space. Let $\mathrm{End}(V)$ denote the vector space of all linear transformations from V to V. Prove that the *commutator bracket* $[f, g] = f \circ g - g \circ f$ defines a Lie Bracket on the vector space $\mathrm{End}(V)$. (Thus, $\mathrm{End}(V)$ equipped with this Lie Bracket is a Lie Algebra, often denoted by $\mathfrak{gl}(V)$ and called the *general linear algebra of* V.)

Proof. Let $\alpha, \beta \in \mathbb{F}$ and let $f, g, h \in \mathrm{End}(V)$.

(i) (Bilinearity) For all $x \in V$, we have

$$\begin{aligned}
[\alpha f + \beta g, h](x) &= (\alpha f + \beta g) \circ h(x) - (h \circ \alpha f + \beta g)(x) \\
g &= (\alpha f + \beta g)(h(x)) - h((\alpha f + \beta g)(x)) \\
&= \alpha f(h(x)) + \beta g(h(x)) - h((\alpha f)(x) + (\beta g)(x)) \\
&= \alpha f(h(x)) + \beta g(h(x)) - h(\alpha f(x)) - h(\beta g(x)) \\
&= \alpha f(h(x)) + \beta g(h(x)) - \alpha h(f(x)) - \beta h(g(x)) \\
&= \alpha (f \circ h)(x) + \beta (g \circ h)(x) - \alpha (h \circ f)(x) - \beta (h \circ g)(x) \\
&= \alpha [f \circ h - h \circ f](x) + \beta [g \circ h - h \circ g](x) \\
&= \alpha [f, h](x) + \beta [g, h](x).
\end{aligned}$$

Hence, we conclude that
$$[\alpha f, \beta g, h] = \alpha [f, h] + \beta [g, h].$$
A similar argument shows bilinearity for the second component.

(ii) (Skew-symmetric) Let $f, g \in \mathrm{End}(V)$. Then

$$[f, g] = f \circ g - g \circ f = -(g \circ f - f \circ g) = -[g, f].$$

(iii) Let $f, g, h \in \mathrm{End}(V)$. Our goal is to show that

$$[f, [g, h]] + [h, [f, g]] + [g, [h, f]] = 0$$

Note that

$$\begin{aligned}
[f, [g, h]] &= f \circ [g, h] - [g, h] \circ f \\
&= f \circ (g \circ h - h \circ g) - (g \circ h - h \circ g) \circ f \\
&= f \circ (g \circ h) - f \circ (h \circ g) - (g \circ h) \circ +(h \circ g) \circ f.
\end{aligned}$$

Similarly, we have that

$$\begin{aligned}
[h, [f, g]] &= h \circ [f, g] - [f, g] \circ h \\
&= h \circ [f \circ g - g \circ f] - [f \circ g - g \circ f] \circ h \\
&= h \circ (f \circ g) - h \circ (g \circ f) - (f \circ g) \circ h + (g \circ f) \circ h
\end{aligned}$$

and

$$\begin{aligned}
[g, [h, f]] &= g \circ [h, f] - [h, f] \circ g \\
&= g \circ [h \circ f - f \circ h] - [h \circ f - f \circ h] \circ g \\
&= g \circ (h \circ f) - g \circ (f \circ h) - (h \circ f) \circ g + (f \circ h) \circ g.
\end{aligned}$$

Then cancelling like terms and collecting, we arrive at

$$[f, [g, h]] + [h, [f, g]] + [g, [h, f]] = 0.$$

Hence, we conclude from above that $[f, g] = f \circ g - g \circ f$ is a Lie Bracket on $\mathrm{End}(V)$. ∎

Problem 10. Let G be any of the matrix Lie groups that we have introduced as embedded submanifolds of $\mathrm{GL}(n, \mathbb{F})$. We know that $T_I G$ is a vector space of matrices. Prove that if $X, Y \in T_I G$, then $[X, Y] = XY - YX \in T_I G$ and hence $[X, Y] = XY - YX$ is a Lie Bracket on $T_I G$. (The space $T_I G$ equipped with this Lie Bracket is the Lie Algebra of G, denoted by $\mathrm{Lie}(G)$ or $\mathfrak{g}$).

Proof. Suppose $X, Y \in T_I G$. Our goal is to show that $[X, Y] \in T_I G$. Since $X, Y \in T_I G$, there exists smooth paths $A(t)$ and $B(t)$ such that

$$A(0) = I \quad \text{and} \quad B(0) = I$$

and

$$A'(0) = X \quad \text{and} \quad B'(0) = Y.$$

Define $C_S(t) = A(s)B(t)A^{-1}(S)$ is also a smooth path in $T_I G$. Hence, we see that for all s, $C_s'(0) = T_I G$. Now, if we let $V(s) = C_s'(0)$, then $V(s)$ is a path inside $T_I G$. Thus, in particular $V'(0) = T_I G$. That is,

$$\begin{aligned}
&\frac{d}{ds}\Big|_{s=0}\left[\frac{\partial}{\partial t}\Big|_{t=0} A(s)B(t)A^{-1}(s)\right] \in T_I G \\
&= \frac{d}{ds}\Big|_{s=0}\left[A(s)B'(0)A^{-1}(s)\right] \\
&= A'(0)B'(0)A^{-1}(0) + A(0)B'(0)(A^{-1}(0))' \\
&= A'(0)B'(0)A^{-1}(0) + A(0)B'(0)[-A(0)A'(0)A^{-1}(0)] \\
&= XYI + IY[-IXI^{-1}] \\
&= XY - YX \\
&= [X, Y].
\end{aligned}$$

Hence, it follows that $[X, Y] \in T_I G$. ∎

Problem 11. Consider the linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ defined by

$$T(x, y, z) = (2x, x + 2y - z, x + 3y - 2z).$$

1. Find the matrix of T with respect to the standard basis $\mathcal{E} = \{e_1, e_2, e_3\}$ of $\mathbb{R}^3$.
2. Verify that the following vectors are eigenvectors of T:

$$v_1 = (1, 1, 1), \quad v_2 = (0, 1, 1) \quad v_3 = (0, 1, 3).$$

3. Let $\mathcal{B} = \{v_1, v_2, v_3\}$ be this basis of eigenvectors. Compute the matrix of T with respect to $\mathcal{B}$ and verify that is diagonal. (**Hint:** Use $[T]_{\mathcal{B}} = P^{-1}[T]_{\mathcal{E}}P$) with the change of basis matrix $P = P_{\mathcal{E}\leftarrow\mathcal{B}} = \begin{pmatrix} v_1 & v_2 & v_3 \end{pmatrix}$.

Solution. 1. Given $e_1 = (1, 0, 0), e_2 = (0, 1, 0), e_3 = (0, 0, 1)$, we have

$$\begin{aligned}
&(x, y, z) = xe_1 + ye_2 + ze_3 \\
&\implies T(x, y, z) = 2xe_1 + (x + 2y - z)e_2 + (x + 3y - 2z)e_3.
\end{aligned}$$

Furthermore, we have

$$\begin{aligned}
T(e_1) &= T(1, 0, 0) = (2, 1, 1) \\
T(e_2) &= T(0, 2, 3) = (0, 2, 3) \\
T(e_3) &= T(0, 0, 1) = (0, -1, -2).
\end{aligned}$$

Using the above, we obtain the matrix representation of T with respect to $\mathcal{E}$:

$$[T]_{\mathcal{E}} = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & -1 \\ 1 & 3 & -2 \end{pmatrix}.$$

∎

Lemma. For any $v \in L$ where L is a Lie Algebra, $[v, v] = 0$.

Proof. Let $v \in L$. By the Skew-symmetric property of $[\cdot, \cdot]$, it follows that

$$[v, v] = -[v, v] \implies 2[v, v] = 0 \implies [v, v] = 0.$$

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Problem 12. A Lie Algebra is said to be *abelian* if

$$\forall x, y \in L \quad [x, y] = 0.$$

Prove that every one-dimensional Lie Algebra is abelian.

Proof. Let L be a one-dimensional Lie Algebra. Our goal is to show that for all $x, y \in L$, $[x, y] = 0$. Let $x, y \in L$ be arbitrary. Since L is a one-dimensional Lie Algebra, L is a one-dimensional vector space. Let $\beta = \{v\}$ be a basis for L. Then $x = \alpha_1 v$ and $y = \alpha_2 v$ for some α_1 and α_2 in $\mathbb{F}$. Using the scaling property of the first and second component of the Lie Bracket, it follows that

$$[x, y] = [\alpha_1 v, \alpha_2 v] = \alpha_1\alpha_2[v, v] = \alpha_1\alpha_2 \cdot 0 = 0.$$

Hence, we see that L is abelian. ■

Problem 13. Recall that a Lie Algebra is a set equipped with three extra structures: addition and scalar multiplication (giving it the structure of a vector space), and a bilinear operation called the Lie Bracket. With this in mind, how should one define a *Lie Algebra Homomorphism* – a structure-preserving map between two Lie Algebras?

Definition (Lie Algebra Homomorphism)**.** Let $\mathfrak{g}$ and $\mathfrak{h}$ be two Lie Algebras over $\mathbb{F}$ with Lie Brackets $[\cdot, \cdot]$ and $[\cdot, \cdot]'$, respectively. We say that the mapping $\varphi : \mathfrak{g} \to \mathfrak{h}$ is a **Lie Algebra Homomorphism** if φ is a linear map (i.e for all $x, y \in \mathfrak{g}$, for all $\alpha \in \mathbb{F}$ $\varphi(\alpha x + y) = \alpha\varphi(x) + \varphi(y)$.) such that

$$\varphi([x, y]) = [\varphi(x), \varphi(y)]' \quad \forall x, y \in \mathfrak{g}.$$

Problem 14. Let L be a real or complex Lie Algebra. For each $x \in L$, define a linear map:

$$\mathrm{ad}_x : L \to L \quad \mathrm{ad}_x(y) = [x, y].$$

1. Show that for every $x \in L$, the map $\mathrm{ad}_x : L \to L$ is linear.

Proof. Let $x \in L$. Then by Bilinearity of $[\cdot, \cdot]$, we have for all $y_1, y_2 \in L$, we have

$$\begin{aligned} \mathrm{ad}_x(y_1 + y_2) &= [x, y_1 + y_2] \\ &= [x, y_1] + [x, y_2] \\ &= \mathrm{ad}_x(y_1) + \mathrm{ad}_{y_2}. \end{aligned}$$

Hence, we have

$$\mathrm{ad}_x(y_1 + y_2) = \mathrm{ad}_x(y_1) + \mathrm{ad}_x(y_2). \tag{1}$$

Now, let $\alpha \in \mathbb{F}$. Then for all $y \in L$, we have

$$\mathrm{ad}_x(\alpha y) = [x, \alpha y] = \alpha[x, y] = \alpha \mathrm{ad}_x(y).$$

Thus,

$$\mathrm{ad}_x(\alpha y) = \alpha \cdot \mathrm{ad}_x(y). \tag{2}$$

Now, it follows from (1) and (2) that $\mathrm{ad}_x : L \to L$ is linear. ■

2. Define $\mathrm{ad} : L \to \mathfrak{gl}(L) \quad x \mapsto \mathrm{ad}_x$. Prove that ad is a linear map and that it preserves the Lie Bracket

$$\forall x, y \in L \quad [\mathrm{ad}_x, \mathrm{ad}_y] = \mathrm{ad}_{[x,y]},$$

where the bracket on the left-hand side is the commutator bracket in $\mathfrak{gl}(L)$, and the bracket on the right-hand side is the Lie Bracket of L.

Proof. We will prove the following properties:

(1) $\mathrm{ad} : L \to \mathfrak{gl}(L)$ defined by $x \mapsto \mathrm{ad}_x$ is linear

(2) $\forall x, y \in L$ $[\mathrm{ad}_x, \mathrm{ad}_y] = \mathrm{ad}_{[x,y]}$.

(1) Let $x, y \in L$. Then for all $z \in L$, we have

$$\begin{aligned}[\mathrm{ad}(x+y)](z) = \mathrm{ad}_{x+y}(z) &= [x+y, z] \\ &= [x,z] + [y,z] \\ &= \mathrm{ad}_x(z) + \mathrm{ad}_y(z).\end{aligned}$$

Then we see that

$$\mathrm{ad}(x+y) = \mathrm{ad}(x) + \mathrm{ad}(y)$$

since $z \in L$ is arbitrary. Now, let $\alpha \in \mathbb{F}$. Then we have for all $z \in L$,

$$\begin{aligned}[\mathrm{ad}(\alpha x)](z) = \mathrm{ad}_{\alpha x}(z) = [\alpha x, z] &= \alpha[x,z] \\ &= \alpha \mathrm{ad}_x(z).\end{aligned}$$

Hence, we see that $\mathrm{ad} : L \to \mathfrak{gl}(L)$ is linear.

(2) Let $x, y \in L$. Then for all $z \in L$, it follows that

$$\begin{aligned}[\mathrm{ad}_x, \mathrm{ad}_y](z) &= (\mathrm{ad}_x \circ \mathrm{ad}_y - \mathrm{ad}_y \circ \mathrm{ad}_x)(z) \\ &= (\mathrm{ad}_x \circ \mathrm{ad}_y)(z) - (\mathrm{ad}_y \circ \mathrm{ad}_x)(z) \\ &= \mathrm{ad}_x([y,z]) - \mathrm{ad}_y([x,z]) \\ &= [x,[y,z]] - [y,[x,z]].\end{aligned}$$

It suffices to show that

$$\mathrm{ad}_{[x,y]}(z) = [[x,y],z] - [[x,y],z].$$

We proceed to use the Jacobi-identity. That is,

$$\begin{aligned}&[y,[x,z]] + [z,[y,x]] + [x,[z,y]] = 0 \\ &\implies [z,[y,x]] + [x,[z,y]] = -[y,[x,z]] \\ &\implies [z,-[x,y]] + [x,[z,y]] = -[y,[x,z]] \\ &\implies -[-[x,y],z] + [x,[z,y]] = -[y,[x,z]] \\ &\implies [[x,y],z] + [x,-[y,z]] = -[y,[x,z]] \\ &\implies [[x,y],z] - [x,[y,z]] = -[y,[x,z]].\end{aligned}$$

Thus, we have that for all $z \in L$,

$$\begin{aligned}[\mathrm{ad}_x, \mathrm{ad}_y]k &= [x,[y,z]] + [[x,y],z] - [x,[y,z]] \\ &= [[x,y],z] \\ &= \mathrm{ad}_{[x,y]}(z)\end{aligned}$$

Thus, we conclude that

$$[\mathrm{ad}_x, \mathrm{ad}_y] = \mathrm{ad}_{[x,y]}(z).$$

■